

MATTHEW BARONDEAU

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<https://matthewbarondeau.github.io/>

EDUCATION

Ph.D. Electrical and Computer Engineering The University of Texas at Austin Dissertation: Efficient Scheduling in Same-ISA Heterogeneous Systems	09/2026
M.S. Electrical and Computer Engineering The University of Texas at Austin; GPA 3.88	12/2022
B.S. Electrical and Computer Engineering The University of Texas at Austin; GPA 3.75	05/2020

WORK EXPERIENCE

Nvidia Resiliency and Safety Architect • TBD • TBD	10/2026 - Present Austin, TX
The University of Texas at Austin Graduate Research Assistant • Researched near-memory and heterogeneous, accelerator-rich architectures • Explored ML-based utilization and modeling improvements for heterogeneous SoCs	08/2020 - 09/2026 Austin, TX
Nvidia Resiliency and Safety Architecture Intern • Standardized and expanded liveness analysis and proposed improvements for GPU datacenter resiliency • Automated hardware safety requirements process for future GPUs	05/2025 - 08/2025 Santa Clara, CA
Nvidia Resiliency and Safety Architecture Intern • Performed liveness analysis and injected faults to improve data center resiliency and automotive safety • Automated process to be chip agnostic for future exploration	05/2024 - 08/2024 Santa Clara, CA
Nvidia GPU Architecture Intern • Enhanced the L2 standalone simulator to simulate inter-slice communication • Designed and optimized a GPU L2 metadata prefetcher	05/2022 - 08/2022 Austin, TX
Tactical Computing Labs Research Engineer I • Quantified the component overhead in SST and developed a performance monitoring tool • Implemented locality optimization for accelerator thread migration, leading to a data movement reduction	05/2021 - 08/2021 Austin, TX
Arm CPU Performance Intern • Established a correlation between RTL and performance simulator replacement policy behavior • Detected and rectified bugs related to the eviction process of the cache replacement policy	05/2020 - 08/2020 Austin, TX
Lockheed Martin Missiles & Fire Control Electrical Engineering Intern • Engineered a high-current power supply controller • Conducted comprehensive testing and characterization of a fiber communication circuit	06/2019 - 08/2019 Grand Prairie, TX
Applied Research Labs Student Technician • Wrote RTL module to enable DMA playback of GPS data on Xilinx Virtex 7 FPGA • Replicated analog front-end attenuation experiment to characterize system noise	05/2018 - 08/2018 Austin, TX

Reynolds & Reynolds

IT Operations Intern

- Repaired company computers, printers, and networking equipment
- Ran ethernet cable and installed new security infrastructure

05/2014 - 08/2017

College Station, TX

TEACHING

The University of Texas at Austin	08/2025 - 12/2025
Compilers Teaching Assistant	Austin, TX
The University of Texas at Austin	08/2021 - 12/2024
First-Year Seminar Instructor	Austin, TX
The University of Texas at Austin	01/2020 - 05/2020, 01/2022 - 05/2022
Real-Time Operating Systems Teaching Assistant	Austin, TX
The University of Texas at Austin	08/2019 - 12/2019
Embedded System Design Lab Teaching Assistant	Austin, TX
The University of Texas at Austin	08/2017 - 12/2019
First-Year Seminar Teaching Assistant	Austin, TX
The University of Texas at Austin	01/2018 - 05/2018, 01/2019 - 05/2019
Introduction to Embedded Systems Teaching Assistant	Austin, TX
The University of Texas at Austin	08/2018 - 12/2018
Introduction to Computing Teaching Assistant	Austin, TX
The University of Texas at Austin	08/2017 - 12/2017
Introduction to Electrical Engineering Teaching Assistant	Austin, TX

SERVICE

Graduate ECE (GREECE), The University of Texas at Austin	01/2024 - 09/2025
Board Member	Austin, TX
ECE Partner Program, The University of Texas at Austin	01/2021 - 09/2025
Graduate Student Peer Mentor	Austin, TX

AWARDS

Texas ECE Graduate Leadership Award	2024
Qualcomm Innovation Fellowship	2023 - 2024
Virginia & Ernest Cockrell Jr. Fellowship in Engineering	2020 - 2024

SKILLS

Languages: C/C++, Python, Perl, CUDA, SystemVerilog, Assembly, HTML, CSS	
Software: Git, Gem5, SST, Perf, Verilator, Intel PIN, Vivado, LLVM, Pandas, Scikit	
Topics: Parallel architecture, GPU architecture, Memory system design, Resilient computing, System modeling	

PUBLICATIONS

- M. Barondeau, S. Jiang, J.C. Beard, and A. Gerstlauer. ViReC: The Virtual Register Context Architecture for Efficient Near-Memory Multithreading. In *International Conference on Parallel Processing*, 2025.
- M. Barondeau, E. Alcorta, M. Koppikar, and A. Gerstlauer. PerfCount: A Performance Counter Dataset for CPU Prediction Problems. In *IEEE International Symposium on Workload Characterization*, 2026.